

Homework Grading Report

Student:	Student_11
Assignment:	Midterm Exam (mt13)
Date:	February 08, 2026
Grade:	103.0 / 125 (82%) B-

Section Breakdown

Section	Points	Earned	Notes
Part 1: R Basics and Data Import	10	10.0	All tasks completed
Part 2: Data Cleaning	15	7.5	Missing: drop_na(), IQR()
Part 3: Basic Data Transformation	15	15.0	Missing: %>%()
Part 4: Advanced Transformation	15	15.0	All tasks completed
Part 5: Data Reshaping	15	15.0	All tasks completed
Part 6: Combining Datasets	15	15.0	All tasks completed
Part 7: String Manipulation & Date/Time	20	20.0	Missing: str_trim()
Part 8: Advanced Wrangling	15	15.0	All tasks completed
Part 9: Reflection Questions	5	5.0	All tasks completed
TOTAL	125	103.0	82%

Detailed Feedback

What You Did Well

- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2)
- Part 2: Data Cleaning - Missing Values & Outliers (Lesson 2)
- Part 3: Data Transformation Part 1 (Lesson 3)
- Part 3: Data Transformation Part 1 (Lesson 3)
- Part 4: Data Transformation Part 2 (Lesson 4)
- Part 4: Data Transformation Part 2 (Lesson 4)
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What to Fix

1. Part 1: R Basics and Data Import (Lesson 1): Your code attempts to set the working directory and load packages, but fails to import datasets due to file path issues. You used `setwd('/workspaces/assignment-1-Chrastianfrone/assignment/Homework')` which may not contain the required CSV files. The error indicates that 'company_sales_data.csv' could not be opened. To fix this, ensure the CSV files are in the specified directory or adjust the path accordingly.

2. Part 6: Combining Datasets with Joins (Lesson 6): Your code creates a duplicate assignment in the line `'orders_with_items <- orders_with_items <- orders %>%'`. This syntax error prevents proper execution. You should remove the redundant assignment and simply write `'orders_with_items <- orders %>% inner_join(order_items, by = "OrderID")'`. This will correctly join the orders and order_items tables on OrderID.

3. To improve missing value handling, you need to assess why values are missing rather than simply dropping them; this matters because dropping rows can bias revenue estimates. Consider using imputation (e.g., median Revenue) with `mutate: sales_data <- sales_data %>% mutate(Revenue = if_else(is.na(Revenue), median(Revenue, na.rm=TRUE), Revenue))`. The solution replaces NAs and preserves row counts.

4. Your outlier detection reports zero outliers, but you only calculated Q1 and Q3 without applying the IQR rule. This matters because undetected extreme values can distort averages. Implement: `lower <- Q1 - 1.5*IQR; upper <- Q3 + 1.5*IQR; sales_clean <- sales_data %>% filter(Revenue >= lower & Revenue <= upper)`. The solution flags any revenue outside this range.

5. The `pivot_longer` and `pivot_wider` steps are missing, which prevents you from creating tidy datasets for analysis. This matters because tidy data enables consistent modeling and visualization. Add: `sales_long <- sales_data %>% pivot_longer(cols = c(Revenue, Units_Sold), names_to = "Metric", values_to = "Value"); sales_wide <- sales_long %>% pivot_wider(names_from = Metric, values_from = Value)`. The solution produces a long table for time-series plots and a wide table for summary tables.

6. Your join operations are not demonstrated; without them you cannot integrate customer, order, and product information. This matters because fragmented data limits insight into customer lifetime value. Use: `orders_full <- customers %>% left_join(orders, by = "CustomerID") %>% left_join(order_items, by = "OrderID")`; the solution yields a dataset with all customer attributes attached to each line item.

7. Reflection answers are brief and lack evidence; this matters because reflective thinking shows you can evaluate the impact of your analytical choices. Expand each answer to 3-4 sentences, include specific numbers from your code (e.g., "After removing 12 outliers, average revenue dropped from \$45,000 to \$38,200"), and discuss business implications.

Reflection & Critical Thinking

You answered Question 9.1 with a generic statement that handling missing values and outliers makes the analysis correct, but you did not explain how the specific cleaning steps you performed (e.g., using `na.omit` and IQR thresholds) changed any summary statistics. Adding a brief comparison of before-and-after totals would demonstrate deeper insight.

In Question 9.2 you identified Europe and Consulting as top performers, which matches your grouped summary, yet you did not discuss why those segments excel or what actions the business could take. Expanding on drivers such as market size or product mix would show stronger analytical thinking.

Your response to Question 9.3 correctly notes that wide format suits dashboards and long format suits calculations, but you did not provide a concrete business scenario (e.g., monthly sales trend analysis vs. quarterly KPI report). Including a specific example would make the explanation more compelling.

For Question 9.4 you correctly distinguished `left_join` and `inner_join`, yet you did not illustrate the impact on your integrated dataset (e.g., number of customers retained vs. lost). Showing a small result table or count would clarify the practical consequences of each join type.

In Question 9.5 you highlighted `mutate` as the most valuable skill, but you did not connect it to a concrete business metric you created (e.g., `revenue_per_unit`). Describing how that new column informed a recommendation would strengthen your argument.

Instructor Assessment

You have shown that you can import data, create basic metrics, and recognize the importance of cleaning and joins, but many required steps are incomplete or only sketched out. Your reflections are too brief to demonstrate full understanding of how your choices affect business insights. Focus on finishing the missing code sections, enrich your written analysis with specific numbers, and connect each analytical result to a clear recommendation for the company.

Key Takeaway

Complete the missing data-reshaping and join sections using the examples provided, then re-run the notebook to ensure all objects exist.

Deepen your reflection responses by linking each analytical step to concrete results and business decisions.

Practice more robust data-quality checks (imputation, outlier flagging) and document the rationale in comments.